Quantum Multimodal Contrastive Learning Framework

From EEG–Image Decoding to Video–Text Retrieval

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Based on Chen, Tsai & Huang, *ICASSP 2025*
Extended to video–text retrieval

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Outline

Problem Statement: The Original Paper

Goal: use quantum computing to improve *multimodal contrastive learning*.

Why is this hard?

- Classical multimodal contrastive learning (CLIP-style) is constrained by classical representation capacity on noisy, high-dimensional signals.
- EEG is non-invasive and high temporal resolution, but low SNR and highly variable – decoding images from EEG is notoriously difficult.
- Quantum encoders (superposition, entanglement) may explore a larger solution space and learn richer cross-modal representations.

Paper task: *zero-shot* EEG $\leftrightarrow$ image recognition using a hybrid quantum–classical contrastive framework (QMCL).

First work to use a Variational Quantum Circuit for self-supervised *cross-modal* contrastive learning.

Problem Statement: This Project's Extension

Research question

Does a quantum encoder still help when the QMCL idea is moved from EEG–image to **video–text retrieval** on top of strong frozen multimodal backbones?

- Reuse the QMCL principle: frozen pretrained encoders + a trainable quantum projection, aligned by a contrastive loss.
- New modality pair: **video** and **natural-language description**.
- Task: bidirectional retrieval – video→text and text→video (Top-1/3/5 accuracy, mean rank, margin).
- Fair comparison against a *classical bottleneck* adapter with the *same parameter budget* – isolating the quantum contribution.

Datasets: Original Paper (EEG–Image)

ThingsEEG (Gifford et al., 2022) – one of the largest EEG–image datasets.

- Rapid Serial Visual Presentation; 10 participants, 16,740 natural images from the THINGS database.
- **1,654 training classes** ($\times$10 images) and **200 test classes** ($\times$1 image) for *zero-shot* evaluation.
- 64-channel EASYCAP; trials segmented 0–1000 ms post-stimulus, baseline corrected, averaged across repetitions; images resized to 224×224.

Datasets: This Project (Video–Text)

Built with a Hugging Face curation + sanity-check pipeline (`curate_hf_video_description_dataset.py`): dedup (exact / perceptual / caption), decode check, duration & length filters.

Manifest	#Items	Notes
`llava-video-0-30-academic`	$\sim$106	long LLaVA-style captions
`finevideo-short-full`	500	mean 76.9 s, 100% unique captions
`msrvtt-short-text-smoke`	smoke	short-caption sanity probe
`qcl-video-text-smoke`	5	pipeline smoke test

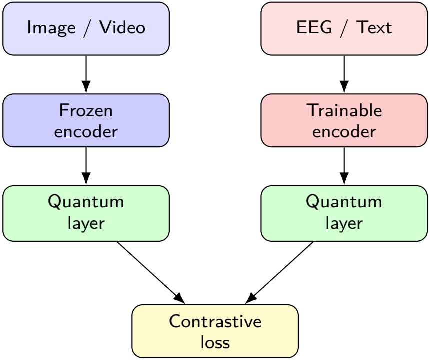
- Main controlled experiment: **500 items** → 350 train / 50 val / 100 test.
- Frozen features cached once (`video_text_feature_cache.pt`) and reused across all adapters for a strictly fair comparison.
- A documented finding: when captions are long/homogeneous and the set is tiny, retrieval is near-random – *data quality dominates*.

Methodology: QMCL Recap

Hybrid quantum–classical two-tower:

1. Classical pre-processing per modality.
2. A *quantum encoding layer* maps classical features into a quantum state.
3. Symmetric CLIP-style contrastive loss aligns the two modalities.

EEG–image instantiation: **frozen** CLIP-ViT image encoder + **trainable** EEG encoder (QSTConv / QNervFormer ± graph attention), each followed by a quantum layer.



Methodology: The Quantum Encoding Layer

Classical features → `Linear` → angle encoding → VQC → measurement → `Linear`.

Variational quantum circuit (PennyLane, 10 qubits, 4 layers)

1. **Input encoding:** per-qubit rotation $R_Y(x_i) = \exp(-i\,\frac{x_i}{2}\sigma_y)$ (`AngleEmbedding`); inputs squashed by $\tanh(\cdot) \cdot \pi$.
2. **Parameterized layers:** `StronglyEntanglingLayers` – trainable single-qubit rotations + a ring of CNOT entanglers, weight shape $(n_{\text{layers}}, n_{\text{qubits}}, 3)$.
3. **Measurement:** $\langle \sigma_z^i \rangle = \langle 0 | U^\dagger \sigma_z^i\, U | 0 \rangle$ for every qubit.

Trained jointly with the classical parts; gradients via the parameter-shift rule (PennyLane Torch interface).

Methodology: Video–Text Pipeline (This Project)

Frozen feature extractors (`VideoTextFeatureExtractor`):

- Video: CLIP ViT-B/32 frame features (mean-pooled) *or* XCLIP (`microsoft/xclip-base-patch32`) chunked video features.
- Text: CLIP / XCLIP text encoder with *chunked* tokenization so long descriptions are not truncated.

Trainable head: residual post-encoder adapter on the video side

$$z = \underbrace{\mathrm{align}(v)}_{\text{frozen feat.}} + \alpha \cdot \mathrm{Adapter}(v), \qquad \alpha \text{ learnable residual scale.}$$

Three interchangeable adapters:

- `raw` – no adapter (frozen-feature baseline).
- `quantum` – `PostEncoderQuantumAdapter` (the VQC above).
- `classical_bottleneck` – Linear $768 \to 10 \to 768$ (**matched parameter budget** control).

Methodology: Matching & Objective

- Multi-chunk videos/captions: cosine similarities over all (video-chunk $\times$ text-chunk) pairs, pooled by **log-mean-exp** (soft-max), or `max`/`mean`.
- Contrastive InfoNCE with a fixed temperature $\tau = 0.07$ (logit scale $= \log(1/\tau)$), not trained for video–text).
- Optimizer AdamW, $\beta = (0.5, 0.999)$; best epoch chosen by validation retrieval loss; final metrics on the held-out test split.
- Identical cached frozen features across adapters $\Rightarrow$ any difference is attributable to the adapter alone.

$$\mathcal{L} \;=\; \mathrm{CE}\Big(\, e^{\log(1/\tau)} \cdot \mathrm{sim}(z_{\text{video}}, z_{\text{text}}),\ \text{labels} \Big)$$

Experimental Setup

Original paper (EEG–image)

- 200-way zero-shot, 10 subjects.
- 4 encoders: QSTConv, QSTConv-GA, QNervFormer, QNervFormer-GA.
- 200 epochs, weighted Adam, lr 2×10^{-4}, 10 qubits / 4 QNN layers, 5 seeds.

This project (video–text)

- 500 items, 350/50/100 split, seed 2024.
- Ablations: `raw` vs `quantum` vs `classical_bottleneck`.
- Backends: CLIP-frame vs XCLIP video encoder.
- 10 qubits / 4 layers, residual scale learnable, $\tau = 0.07$.

Quantum platform

All circuits executed on the PennyLane `default.qubit` **state-vector simulator** (analytic, noise-free).

Results: Original Paper (ThingsEEG, 200-way zero-shot)

Average accuracy over 10 subjects (5 seeds):

Model	Avg Top-1	Avg Top-5
QSTConv	3.5%	13.7%
QSTConv-GA	4.0%	16.8%
QNervFormer	5.1%	20.2%
QNervFormer-GA	**5.2%**	**20.3%**

- All models far above the 0.5% random baseline (200 classes).
- Graph attention + cross-attention (QNervFormer-GA) generalizes best; quantum encoders are competitive on a very hard zero-shot task.

Results: The Data-Quality Turning Point

First attempts (poor results).

- Initial data: LLaVA-style long, homogeneous paragraph captions on tiny galleries; an MSR-VTT smoke pull *rejected all* 200 rows (`accepted: 0`, sanity `pass: false`).
- Symptom: on a 13-item long-caption XCLIP set, `quantum`, `classical_bottleneck` and `identity` all collapsed to **near-random** ($\approx$0.08 Top-1) – no adapter could help.

Fix: switch to a curated FineVideo set.

- Built with the HF curation pipeline (`build_hf_video_text_dataset.py`): FineVideo clip/caption mining + exact (SHA-256) and perceptual near-duplicate video dedup + caption-uniqueness gate.
- `finevideo-short-full`: 500 items, **100% unique captions**, 0.0006 duplicate rate, sanity `pass: true`.

Outcome

Only after the data switch did retrieval become meaningful, and the quantum adapter then showed a clear, reproducible gain (next slide).

Results: Video–Text (Simulator, Main Controlled Run)

Curated FineVideo set, 500 items, 100-item test split, seed 2024:

Adapter	V→T Top-1	V→T Top-5	T→V Top-1	Margin
`raw` (frozen)	0.62	0.86	0.69	1.30
`classical_bottleneck`	0.73	**0.93**	0.63	2.76
`quantum`	**0.76**	0.92	**0.70**	**2.34**

- Quantum adapter: **+0.14** Top-1 and **+0.06** Top-5 video→text over the frozen baseline.
- Contrastive margin nearly doubles ($1.30 \to 2.34$): better-separated embeddings, and mean rank improves $4.74 \to 3.30$.
- A same-budget *classical* bottleneck reaches comparable accuracy , the gain is largely the trainable residual head, *not* uniquely quantum.

Results: Stress Tests

- **Pre-curation data (recap):** on the tiny long-caption set all adapters were near-random, suggesting a *data-quality*, not adapter, bottleneck (see previous turning-point slide).
- **Tiny synthetic probe (64-item gallery):** quantum ($\approx$0.03 Top-1) underperforms classical ($\approx$0.13) when data is scarce and noisy, so quantum advantage is not free.

Takeaway

The quantum adapter helps on a clean, well-curated set; gains shrink or vanish under data scarcity, and a hardware-noise gap remains untested.

Conclusion: Lessons Learned

1. **QMCL transfers across modalities.** The frozen-encoder + quantum-residual recipe improves video→text retrieval (+0.14 Top-1) on a controlled set.
2. **Always include a parameter-matched classical control.** A linear bottleneck of equal size matches the quantum adapter – without it, the "quantum gain" would be over-claimed.
3. **Data quality dominates.** Small, duplicated, or over-long-caption datasets drive metrics to random; a curation + sanity-check stage was decisive.
4. **Fair protocol matters.** Shared cached features, fixed seeds, and best-epoch-by-validation make adapter comparisons trustworthy.

Future work: larger curated corpora, quantum text adapter, noisy-simulator and real-device runs, more qubits/layers.

Thank you!

Questions & Discussion

Base paper: Chen, Tsai & Huang, *Quantum Multimodal Contrastive Learning Framework*, ICASSP 2025
arXiv:2408.13919